Customer Segmentation Analysis Using K-Means Clustering

1. Introduction

Customer behavior is different for every business. Some customers purchase regularly, while others purchase only occasionally.

In this project, I analyzed customer transaction data to understand these differences. I used RFM analysis and K-Means clustering to group customers based on their purchasing behavior.

2. Objective

The main objectives of this project were to:

- Understand customer purchasing behavior.
- Clean the transaction data.
- Calculate RFM values.
- Group customers using K-Means clustering.
- Understand the different customer groups.
- Suggest marketing strategies for different groups.

3. Dataset

I used the Online Retail Dataset for this project.

It contains information about customer transactions, including product details, quantity purchased, transaction date, price, customer ID, and country.

The main columns used for the analysis were CustomerID, InvoiceNo, InvoiceDate, Quantity, and UnitPrice.

4. Tools Used

The following tools were used:

- Python
- Pandas
- NumPy
- Matplotlib

- Seaborn
- Scikit-learn
- Jupyter Notebook

5. Data Cleaning

- Before starting the analysis, I checked the dataset for missing values, duplicate records, and incorrect values.
- I removed records where CustomerID was missing because customer information was required for the analysis.
- I also removed duplicate records and transactions with zero or negative quantities and prices.
- The InvoiceDate column was converted into the correct date format.
- A new column called TotalAmount was created using:
  TotalAmount = Quantity × UnitPrice

6. RFM Analysis

I used RFM analysis to understand customer behavior.

Recency shows how recently a customer made a purchase.

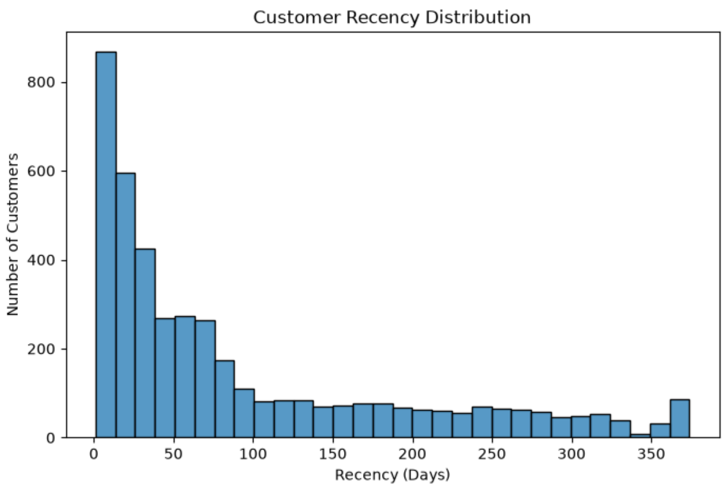
Frequency shows how often a customer made purchases.

Monetary shows the total amount spent by a customer.

These three values were calculated for each customer and used for segmentation.

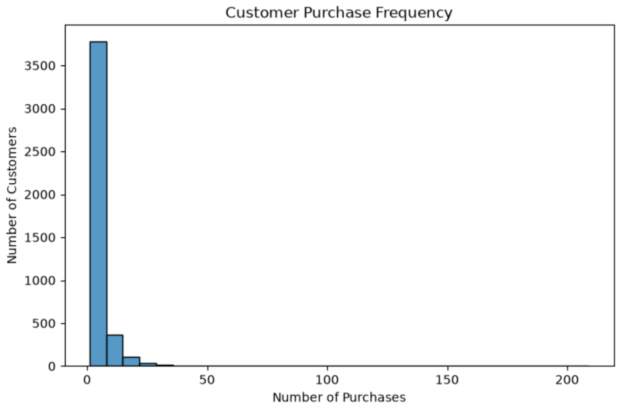
7. RFM Visualization

Recency Distribution



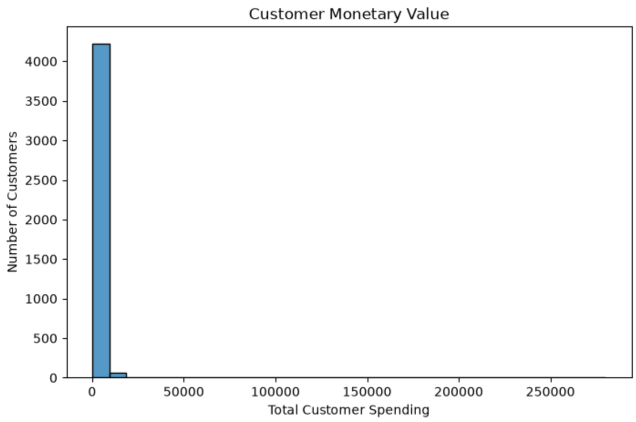
The graph shows how recently customers made their purchases. Customers with lower Recency values are more recently active.

Frequency Distribution



The graph shows that customers have different purchasing frequencies. Some customers purchase much more frequently than others.

Monetary Distribution



The graph shows differences in customer spending. Some customers contribute much more revenue than others.

8. Standardization

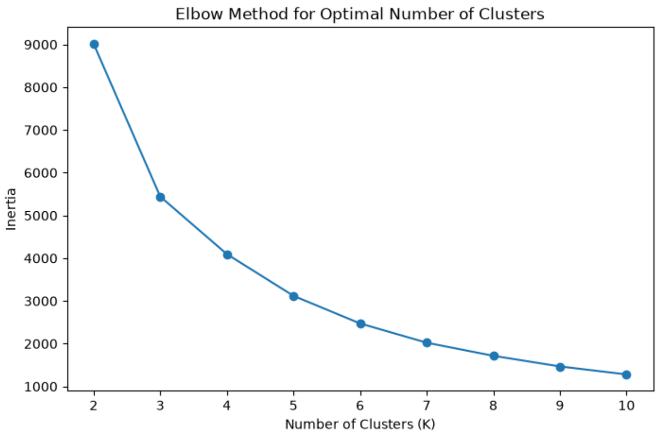
The RFM values had different scales, so I used StandardScaler to standardize them before applying K-Means clustering.

This helped make the three features comparable during the clustering process.

9. Elbow Method

I used the Elbow Method to determine the appropriate number of clusters.

I tested different values of K from 2 to 10.



Based on the graph, I selected 4 clusters for the final analysis.

10. K-Means Clustering

After selecting four clusters, I applied the K-Means algorithm.

Each customer was assigned to one of the four clusters based on their RFM values.

This helped divide the customers into groups with similar purchasing behavior.

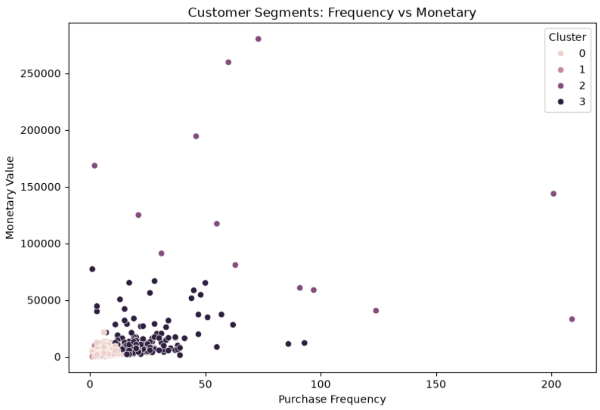
11. Cluster Analysis

I calculated the average Recency, Frequency, and Monetary values for each cluster.

The clusters were then studied to understand the type of customers present in each group.

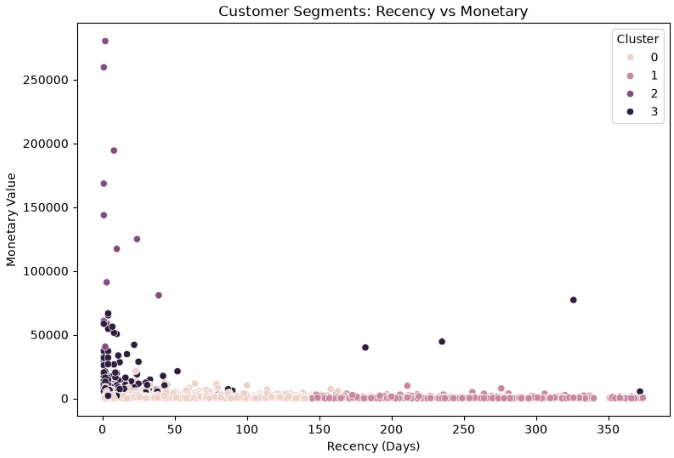
12. Cluster Visualization

Frequency vs Monetary



This graph shows the relationship between how frequently customers purchase and how much they spend.

Recency vs Monetary



This graph helps identify customers who spend more but may not have purchased recently.

13. Marketing Recommendations

Different customer groups can be approached with different marketing strategies.

- **High-value customers:** Offer loyalty rewards, exclusive discounts, and personalized recommendations.
- **Loyal customers:** Provide loyalty benefits and offers that encourage repeat purchases.
- **At-risk customers:** Use re-engagement emails, special discounts, and win-back offers.
- **Low-value customers:** Use simple promotions, product recommendations, and bundle offers.

The exact customer type for each cluster should be based on the RFM values from the analysis.

14. Key Insights

- The analysis showed that customers have different purchasing patterns.
- Some customers purchase frequently and spend more, while others purchase less often.
- RFM analysis helped summarize these differences, while K-Means clustering helped group customers with similar behavior.

- This segmentation can help businesses focus their marketing efforts on the right customers.

15. Conclusion

This project helped me understand how customer transaction data can be used for customer segmentation.

By using RFM analysis and K-Means clustering, I divided customers into four groups based on their purchasing behavior.

The results can help businesses create more personalized marketing campaigns, improve customer engagement, retain valuable customers, and make better marketing decisions.

16. Future Scope

This project can be extended by:

- Creating a Power BI dashboard.
- Predicting customer churn.
- Building a recommendation system.
- Comparing K-Means with other clustering methods.
- Tracking customer segments over time.